

Course Title	Electrical Network Analysis	Course Code	EE2004
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	Credit Hrs.	3 + 1
Knowledge Area	Electrical Engineering Fundamental (KA06)	Grading Scheme	Relative
HEC Knowledge Area	Engineering Foundation	Applicable From	Fall 2023
Pre-requisite(s)	EE1001, EE1001		

Course Objective	The objective of this course is to apply linear circuit analysis tools to solve AC circuits using phasor-domain and Laplace transform analysis techniques.
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No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Apply phasor-domain analysis to solve circuits containing R, L, C, and mutual inductance	A1, Q1, M1, F	C3	1
2	Construct power triangle to compute power in AC circuits.	A2, Q2, M2, F	C3	2
3	Analyze balanced three-phase circuits.	Q3, M2, F	C4	1
4	Apply Laplace transform to analyze RLC circuits.	A3, F	C3	2
5	Analyze various types of filters using s-domain analytics.	Q4, F	C4	2

Text Book(s)	Title	Electric Circuits, 10th Edition
	Author	James W. Nilsson and Susan A. Riedel
	Publisher	Prentice Hall
Ref. Book(s)	Title	Basic Engineering Circuit Analysis, 11th Edition
	Author	J. David Irwin, R. Mark Nelms
	Publisher	

Week	Course Contents/Topics	Chapter*	CLO
1	Introduction to course, The Sinusoidal Source and Response	9	1
2	Phasor, Passive Elements in Frequency Domain	9	1
3	Kirchhoff's Laws in Frequency domain, Series-parallel simplifications, Node and Mesh Analysis in frequency domain	9	1
4	Source Transformation and Thevenin Equivalent in frequency domain, Phasor Diagrams	9	1
5	Mutual Inductance, Transformers	6,4, 9	1
6	Instantaneous, Average, and Reactive Powers, RMS value, Complex Power	10	2
7	Power calculations, Maximum power transfer	10	2
8	Maximum power transfer with restrictions, Introduction to balanced 3-phase circuits	10, 11	2, 3
9	Analysis of Y-Y circuit, Analysis of Y-Δ circuit, Power in 3-phase circuits	11	3
10	Power in 3-phase circuits; Laplace Transform definition, Step Function, Impulse Function	11, 12	3, 4
11	Functional and Operational Transforms, Inverse Transforms	12	4
12	Inverse Transforms, Initial and Final-Value Theorems, Circuit Analysis in s-domain	12, 13	4
13	Application of Laplace Transform to determine the natural and transient response, The Transfer Function	13	4
14	Filters basics, Low-pass and High pass filters	14	5
15	Band-pass and Band-reject filters	14	5

*Reference book chapters are given in brackets

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterm (I+II)	30.0%
Final Exam	50.0%